

AI Assisted Coding

Assignment-7.4

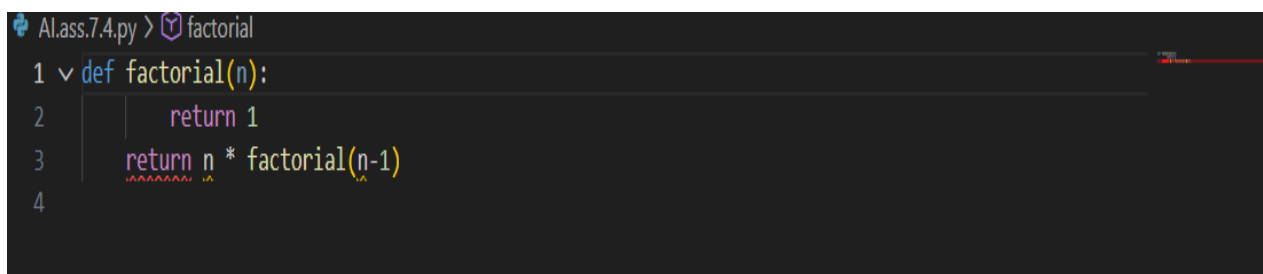
2303A510A0(Vishnu Vardhan)

[Batch-14]

#Error Debugging with AI – Systematic Approaches to Finding and Fixing Bugs

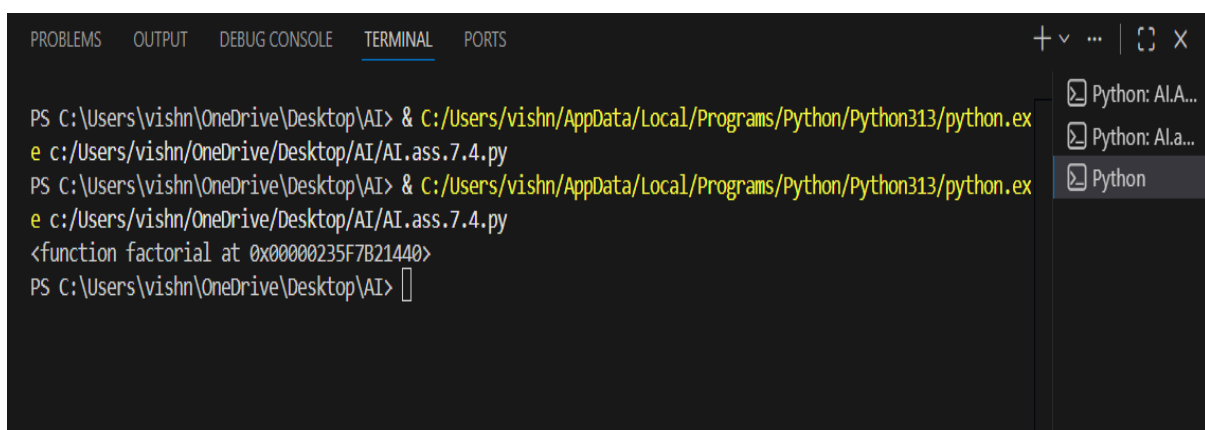
Task-1: Debugging a Recursive Calculation Module

#Error Code:



```
AI.ass.7.4.py > factorial
1 def factorial(n):
2     return 1
3     return n * factorial(n-1)
4
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe
e c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe
e c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py
<function factorial at 0x00000235F7B21440>
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

Explanation:

CHAT

← IDENTIFYING THE ERROR IN CODE

what is the error, fix it

✓ Created inspection plan and read file

I'll create a short plan, then open `Al.ass.7.4.py` to locate the error.

✓ Managed and tracked todo items for planning

Now I'll patch `Al.ass.7.4.py` to add a proper base case and call the function.

✓ Edited `Al.ass.7.4.py` +0 -0

 Run `pylanceSyntaxErrors` – pylance mcp server (MCP Server) 

Validate Python code snippets for syntax errors without saving to file. Returns syntax error details with line numbers and descriptions. Use for: validating generated code, [Show More](#)

Input

```
{ "code": "def factorial(n):\n    if n < 0:\n        raise ValueEr
```

See more

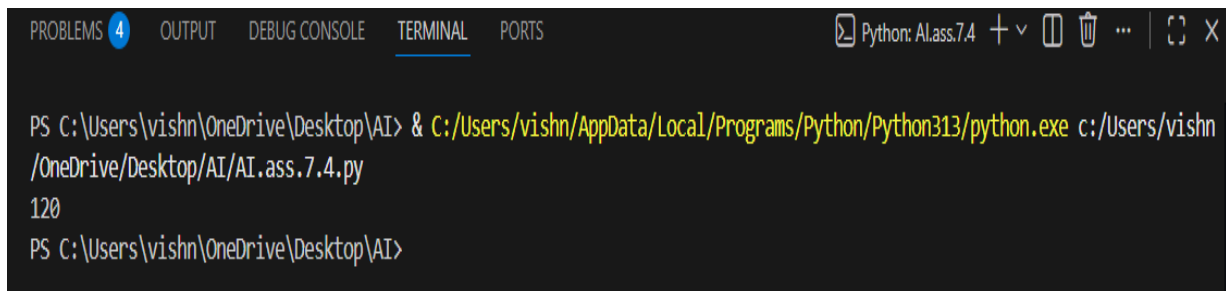
Note that MCP servers or malicious conversation content may attempt to misuse 'Code' through tools.

Allow |  Skip

Fixed code:

```
Al.ass.7.4.py •  
Al.ass.7.4.py > factorial  
1 def factorial(n):  
2     if n < 0:  
3         raise ValueError("n must be non-negative")  
4     if n == 0 or n == 1:  
5         return 1  
6     return n * factorial(n - 1)  
7  
8 print(factorial)
```

Output:

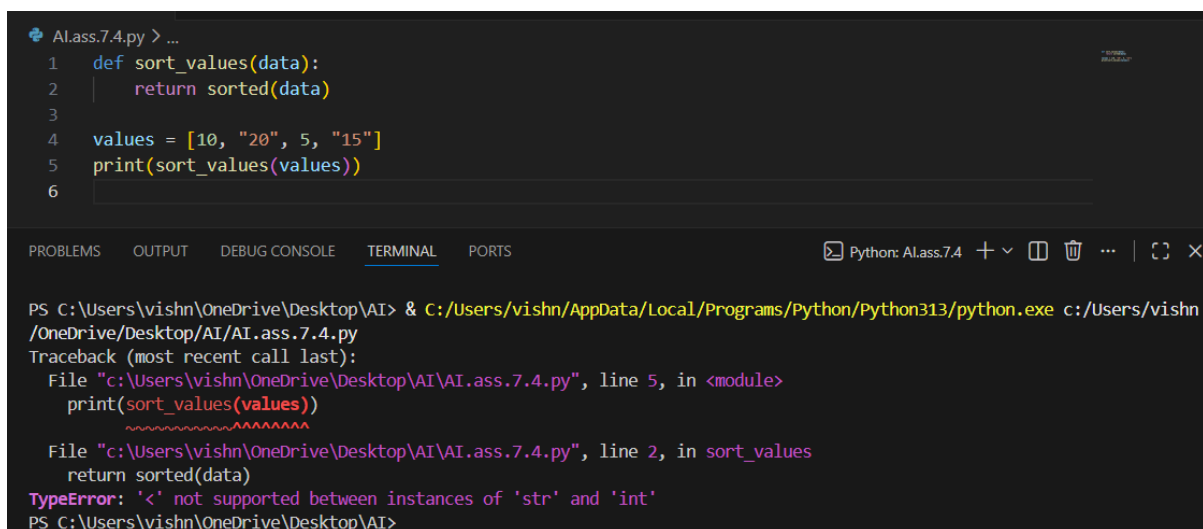


```
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS Python: Al.ass.7.4 + - [ ] [ ] ... [ ] [ ] X
```

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py
120
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

Task-2: Fixing Data Type Errors in a Sorting Utility

#ErrorCode& Output:



```
Al.ass.7.4.py > ...
1 def sort_values(data):
2     return sorted(data)
3
4 values = [10, "20", 5, "15"]
5 print(sort_values(values))
6
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python: Al.ass.7.4 + - [ ] [ ] ... [ ] [ ] X
```

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py
Traceback (most recent call last):
  File "c:\Users\vishn\OneDrive\Desktop\AI\AI.ass.7.4.py", line 5, in <module>
    print(sort_values(values))
    ~~~~~~^~~~~~
  File "c:\Users\vishn\OneDrive\Desktop\AI\AI.ass.7.4.py", line 2, in sort_values
    return sorted(data)
TypeError: '<' not supported between instances of 'str' and 'int'
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

Explanation:

explain and fix the error

✓ Read Python file

Error: Attempting to sort a list with mixed types (integers and strings). Python 3 doesn't allow comparison between `int` and `str`, causing a `TypeError`.

Fix: Convert all values to the same type (integers) before sorting.

✓ Edited `AI.ass.7.4.py` +0 -0

Fixed. The script now converts all values to integers in the sorted function, producing `[5, 10, 15, 20]`.

Claude Haiku 4.5 • 1x

Fixing Error:

```
AI.ass.7.4.py > ...
1 def sort_values(data):
2     return sorted(data)
3     return sorted(int(x) for x in data)
4 values = [10, "20", 5, "15"]
5 print(sort_values(values))
```

Fixedcode with output:

```
AI.ass.7.4.py > ...
1 def sort_values(data):
2     return sorted(int(x) for x in data)
3
4 values = [10, "20", 5, "15"]
5 print(sort_values(values))
6
```

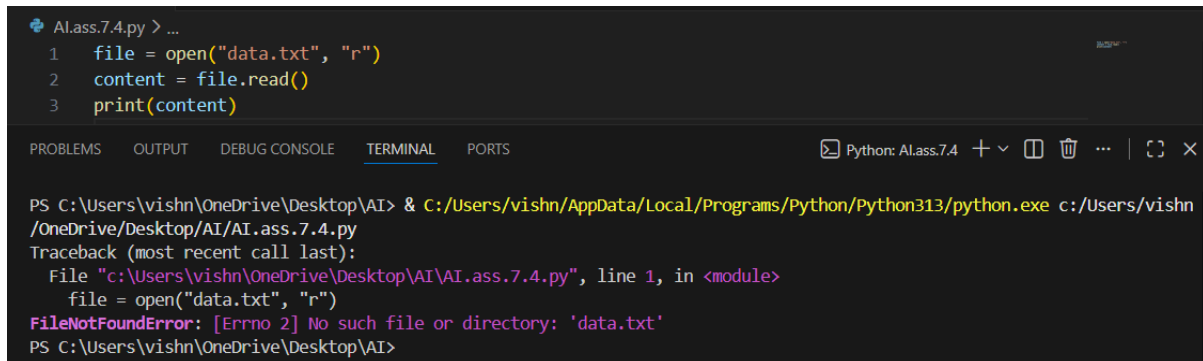
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python: AI.ass.7.4 + ▾ ... |

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py
[5, 10, 15, 20]
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

Task-3: Improving File Handling Reliability

ErrorCode with Output:



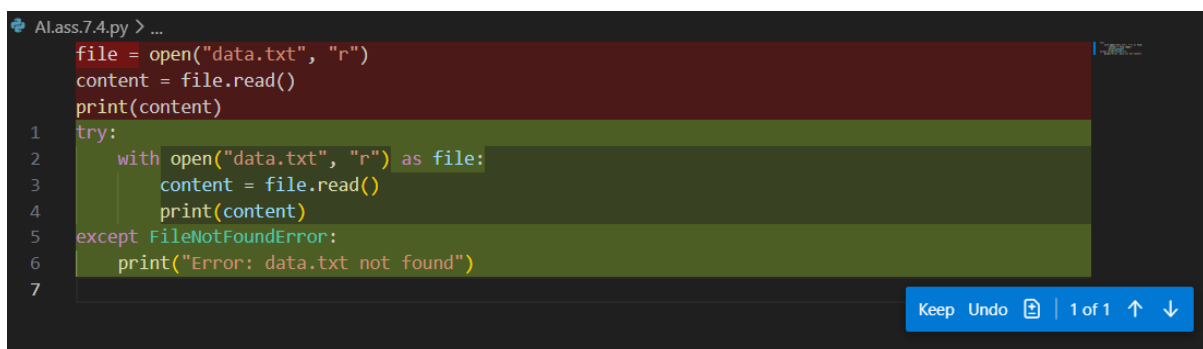
The screenshot shows a code editor with a file named `AI.ass.7.4.py`. The code is as follows:

```
1 file = open("data.txt", "r")
2 content = file.read()
3 print(content)
```

Below the code, the terminal output is displayed. It shows the command to run the script and the resulting error message:

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py
Traceback (most recent call last):
  File "c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py", line 1, in <module>
    file = open("data.txt", "r")
FileNotFoundError: [Errno 2] No such file or directory: 'data.txt'
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

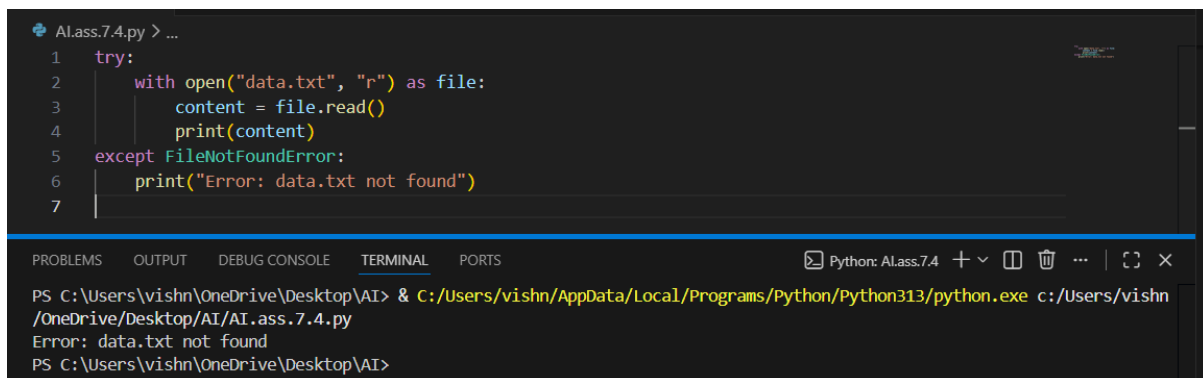
Fixing Error:



The screenshot shows the same code editor with the corrected script:

```
1 try:
2     with open("data.txt", "r") as file:
3         content = file.read()
4         print(content)
5 except FileNotFoundError:
6     print("Error: data.txt not found")
7
```

Fixedcode with output:



The screenshot shows the code editor with the corrected script and the terminal output. The script now handles the missing file gracefully:

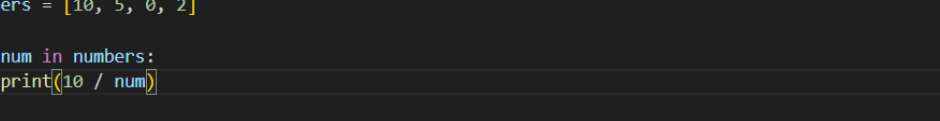
```
1 try:
2     with open("data.txt", "r") as file:
3         content = file.read()
4         print(content)
5 except FileNotFoundError:
6     print("Error: data.txt not found")
7
```

The terminal output shows the command and the successful execution of the script, followed by the error message:

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py
Error: data.txt not found
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

Task-4: Handling Runtime Errors Gracefully in Loops

#ErrorCode& Output:



```
AI.ass.7.4.py > ...
1  numbers = [10, 5, 0, 2]
2
3  for num in numbers:
4      print(10 / num)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python: AI.ass.7.4

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py
1.0
2.0
Traceback (most recent call last):
  File "c:\Users\vishn\OneDrive\Desktop\AI\AI.ass.7.4.py", line 4, in <module>
    print(10 / num)
          ~~~~
ZeroDivisionError: division by zero
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

Fixing Error:

```
Al.ass.7.4.py > ...
1 numbers = [10, 5, 0, 2]
2
3 for num in numbers:
4     print(10 / num)
5     try:
6         print(10 / num)
7     except ZeroDivisionError:
8         print(f"Error: Cannot divide by zero (num={num})")
```

Fixedcode with output:

```
AI.ass.7.4.py > ...
1 numbers = [10, 5, 0, 2]
2
3 for num in numbers:
4     try:
5         print(10 / num)
6     except ZeroDivisionError:
7         print(f"Error: Cannot divide by zero (num={num})")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python: Alass.7.4 + - [] [X] [Y]

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:\Users\vishn\AppData\Local\Programs\Python\Python313/python.exe c:\Users\vishn\OneDrive\Desktop\AI\AI.ass.7.4.py
1.0
2.0
Error: Cannot divide by zero (num=0)
5.0
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

Task-5: Debugging Class Initialization Errors

ErrorCode with Output:

```
AI.ass.7.4.py > ...
1 class Student:
2     def __init__(name, age):
3         name = name
4         age = age
5
6     def display():
7         print(name, age)
8
9 s = Student("Vishnu", 20)
10 s.display()
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS Python: AI.ass.7.4

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py
Traceback (most recent call last):
  File "c:\Users\vishn\OneDrive\Desktop\AI\AI.ass.7.4.py", line 9, in <module>
    s = Student("Vishnu", 20)
TypeError: Student.__init__() takes 2 positional arguments but 3 were given
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

Explanation of original errors and fixes:

Original errors: `__init__` and `display` omitted the `self` parameter; assignments like `name = name` only created local names and `name/age` referenced in `display` were undefined, causing `TypeError/NameError`.

Fixes applied: Changed `def __init__(self, name, age):` and `def display(self):`, and assigned `self.name = name`, `self.age = age`. This ensures instance attributes are stored and accessible.

Fixing Error:

```
Al.ass.7.4.py > ...
1  class Student:
2      def __init__(name, age):
3          name = name
4          age = age
5      """Simple Student class demonstrating proper use of self."""
6
7      def __init__(self, name, age):
8          self.name = name
9          self.age = age
10
11      def display():
12          print(name, age)
13      def display(self):
14          print(self.name, self.age)
15
16 s = Student("Vishnu", 20)
17 s.display()
```

Keep Undo 2 of 2

Fixedcode with output:

```
Al.ass.7.4.py X
Al.ass.7.4.py > Student
1  class Student:
2      def __init__(self, name, age):
3          self.name = name
4          self.age = age
5
6      def display(self):
7          print(self.name, self.age)
8
9
10 s = Student("Vishnu", 20)
11 s.display()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python: Al.ass.7.4

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c:/Users/vishn/OneDrive/Desktop/AI/AI.ass.7.4.py
Vishnu 20
PS C:\Users\vishn\OneDrive\Desktop\AI>
```